An Automated Vision-Based System for Industrial Defect Detection and Localization

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I. Introduction

Industrial manufacturing requires reliable quality inspection to identify defects such as scratches, cracks, dents, contamination, and surface irregularities before products reach customers. Conventional manual inspection is time-consuming, subjective, and difficult to scale for high-speed production lines. Traditional image-processing methods also depend on handcrafted features and fixed rules, making them less robust to changes in lighting, texture, product appearance, and defect characteristics. Recent advances in computer vision and deep learning have enabled automated visual inspection systems that learn discriminative features directly from product images and perform defect classification, detection, and localization [1]–[4].

However, industrial defect datasets commonly contain few defective samples because manufacturing processes are designed to minimize defects. Obtaining detailed annotations for rare defects is also costly, limiting fully supervised models [5]–[8]. This has motivated research into semi-supervised, few-shot, self-supervised, and anomaly-detection approaches that can learn from limited defect information [9], [10]. The proposed project aims to develop an automated industrial defect detection and localization system using deep learning-based computer vision techniques. The system will be evaluated on the MVTec Anomaly Detection dataset, focusing on anomaly detection, defect localization, generalization, and computational efficiency.

II. Literature Review

Recent research has demonstrated a significant shift from traditional handcrafted image-processing techniques toward deep learning-based approaches for industrial visual inspection. Convolutional neural networks, encoder-decoder architectures, attention mechanisms, transformer-based models, and generative approaches have been investigated for defect classification, detection, segmentation, and anomaly detection. These approaches enable models to learn discriminative visual representations directly from industrial images, reducing dependence on manually designed features.

However, the availability of labeled defective samples remains a major challenge in industrial environments. Manufacturing defects are relatively rare, and collecting and annotating sufficient defective samples can be expensive and time-consuming. Consequently, researchers have explored mixed-supervision, few-shot, and self-supervised learning approaches to reduce dependence on large labeled datasets.

Another important research direction is accurate defect detection and localization, particularly for small, irregular, and multi-scale defects. Recent methods have investigated improved object-detection architectures, transformer-based models, feature-enhancement techniques, and generative approaches to improve detection accuracy and localization performance.

Despite these developments, deploying defect detection models in practical manufacturing environments remains challenging. Variations in imaging conditions, product appearance, and manufacturing processes can cause domain shifts between training and deployment data. Recent studies have therefore investigated domain adaptation, transferable representations, lightweight architectures, and efficient anomaly-detection techniques to improve generalization and computational efficiency.

III. Research Gaps

A. Limited defective samples

Industrial defects are rare by nature, meaning collecting thousands of labeled defect images is impractical. Existing supervised approaches can therefore struggle when only a small number of defective samples are available.

B. Unseen defects

Many supervised models learn predefined defect categories. When a new defect type appears, the model may fail to recognize it. This motivates anomaly-detection approaches that learn the characteristics of normal products instead.

C. Small and irregular defects

Small defects can be difficult to distinguish from the surrounding surface, resulting in missed detections or inaccurate localization. This remains a significant challenge in industrial inspection.

D. Accuracy versus real-time performance

Highly accurate models can require substantial computational resources. Industrial systems require both reliable detection and low inference latency, creating a trade-off between accuracy and deployment efficiency.

Thus, there is a need for an industrial defect detection system that can achieve reliable defect detection and localization despite limited defective training samples, small or irregular defects, variations in industrial imaging conditions, and practical computational constraints.

IV. Objectives

a) Primary Objective:: To design, implement, and benchmark an explainable, multi-paradigm automated visual quality inspection system for industrial surface defect classification and anomaly detection.

b) Specific Objectives::

1) Dataset Pipeline & Preprocessing:
   - Implement an automated preprocessing pipeline incorporating noise reduction, Contrast Limited Adaptive Histogram Equalization (CLAHE), and dynamic Region of Interest (ROI) extraction.
   - Design an aggressive data augmentation strategy (rotations, elastic distortion, Gaussian noise, brightness/contrast jitter) using Albumentations.
2) Classical CV Baseline Development:
   - Engineer hand-crafted feature extractors based on Canny edge density distributions, intensity histograms, and GLCM statistical texture matrices.
   - Train and evaluate a Random Forest baseline classifier to establish an empirical benchmark.
3) Supervised Deep Classification:
   - Fine-tune a pretrained EfficientNet-B0 architecture using transfer learning and learning rate scheduling (Cosine Annealing with AdamW).
   - Evaluate multi-class classification performance using Confusion Matrices, Precision, Recall (targeting $\geq 95\%$ defect recall), F1-score, and macro-averaged ROC-AUC.
4) Unsupervised Anomaly Detection Integration:
   - Implement the PatchCore anomaly detection framework utilizing pretrained feature embeddings and coreset memory banks.
   - Demonstrate the ability to detect anomalous samples trained exclusively on non-defective images, evaluated via AUROC.
5) Explainability & Verification: Integrate Grad-CAM to generate pixel-level activation heatmaps overlaid on input inspection images for visual defect localization and verification.
6) Interactive Deployment & Logging: Develop an intuitive Streamlit dashboard enabling single-image and batch image inspection, real-time PASS/FAIL verdict display, confidence distribution charts, and persistent audit logging to CSV.

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